Calculus II **Worksheet** Name:

1. Graph the function $r = 3(1 - \sin(\theta))$.

2. Set up an integral to find the area enclosed by $r = 3(1 - \sin(\theta))$.

3. Find the area enclosed by three leaves of the four-leafed rose $r = 5\cos(2\theta)$.

4. Evaluate $\int \cos^2(3x)\, dx$.

5. Set up an integral to find the length of the curve $y = 4\ln x$, from $x = 1$ to $x = 3$
   (a) By integrating with respect to x.

   (b) By integrating with respect to y.

6. Set up an integral to find the area of the surface generated by rotating the curve $y = 4\ln x$, from $x = 1$ to $x = 3$
   (a) About the x-axis.

   (b) About the line $y = -3$.

   (c) About the line $x = -3$.